

Marking Scheme Total marks = 50

Multiple Choice questions: Each question is worth 1 mark

1. As a mine is drilled deeper and deeper into the Earth, conditions become:
 - a) Cooler
 - b) Wetter
 - c) Hotter**
 - d) Smellier
2. Which of the following is a list of the three rock types?
 - a) igneous, sedimentary, magma
 - b) igneous, hydrous, metamorphic
 - c) sedimentary, metamorphic, sediments
 - d) igneous, sedimentary, metamorphic**
3. Rocks deep down in the mantle are very hot, but they are not totally melted because:
 - a) The temperature is still too low for rocks to melt
 - b) They are also under very high pressure**
 - c) They have a different crystal structure
 - d) They don't contain any water
4. Sandstone is a rock that contains small grains of sand that are cemented together. These grains of sand probably came from:
 - a) Other rocks that were broken down**
 - b) A volcano
 - c) Crystals that formed in the ocean
 - d) Meteorites and asteroids that have hit the Earth
5. Which one of the following is NOT a mineral?
 - a) Quartz
 - b) Granite**
 - c) Biotite
 - d) Feldspar

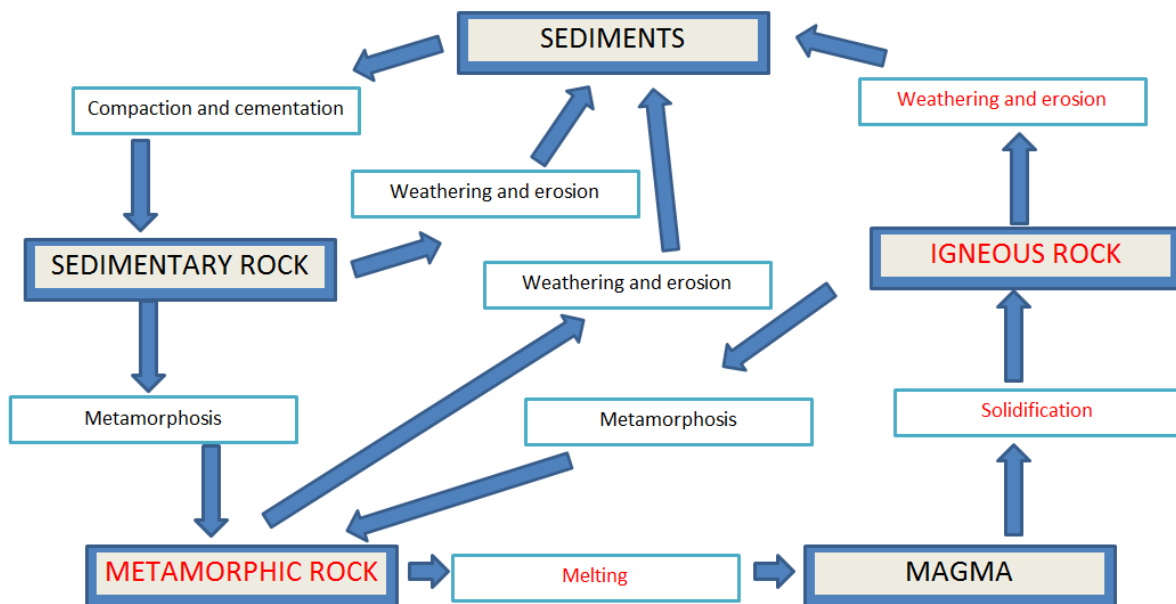
6. A rock contains large crystals that appear to have grown together and are tightly interlinked. The rock is most likely;
 - a) A volcanic igneous rock
 - b) A biogenic rock
 - c) A sedimentary rock
 - d) A plutonic igneous rock
7. The Australian landscape has many granite boulders. Granites are rocks that form deep beneath the Earth. They are lying on the surface today because;
 - a) Volcanoes have blasted them to the surface.
 - b) Weathering and erosion have worn away the rocks above them.
 - c) Glaciers have carried them from a long distance away.
 - d) They are left over from mining activities.
8. A sedimentary rock made from pebbles and gravel cemented together is;
 - a) Sandstone
 - b) Mudstone
 - c) Conglomerate
 - d) Limestone
9. The sedimentary rock in question 8 is likely to have formed from sediments from:
 - a) A fast flowing stream or river
 - b) A river estuary (where the river meets the sea or ocean)
 - c) The shallow ocean
 - d) A desert
10. Metamorphic rocks are usually created;
 - a) During a volcanic eruption
 - b) As rocks experience high temperatures and pressures deep underground
 - c) At the bottom of very deep oceans
 - d) In glaciers
11. The study of large beds of sedimentary rocks has shown that:
 - a) The sediments are laid down horizontally with the oldest sediments at the bottom
 - b) The sediments are laid down horizontally with the youngest sediments at the bottom
 - c) The sediments are laid down vertically with the oldest sediments at the left
 - d) The sediments are laid down vertically with the youngest sediments at the bottom

12. The rock cycle describes how;
- a) New rocks can form from molten rock that solidifies
 - b) New rocks can form as rocks experience extreme temperatures and pressures
 - c) New rocks can form as sediments are compacted deeper and deeper under other sediments
 - d) All of the above
13. The most important, and profitable, mineral mined in Australia is;
- a) diamond
 - b) gold
 - c) opal
 - d) haematite
14. A rock that contains fossils is most probably;
- a) An igneous rock
 - b) A metamorphic rock
 - c) A sedimentary rock
 - d) An igneous or a metamorphic rock

Short Answer Questions (Marks are indicated on each question)

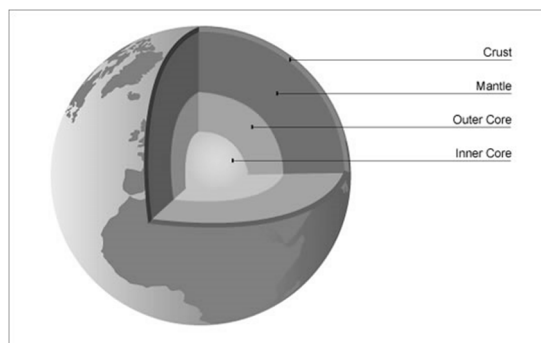
- 15 Fill in the empty spaces on the rock cycle diagram using these terms. **(5 marks)**

igneous rocks, metamorphic rocks, melting, weathering and erosion, solidification



- 16 Label the four layers indicated on this diagram.

(4 marks)



- 17 Toasters often have a layer of a mineral to support the element. This mineral is used because it is extremely heat resistant. In appearance it has flat sheets that can be flaked off, is quite soft, and has a silvery colour.

(3 marks)

Answer: Muscovite

- 18 The rock cycle describes how rocks break down and how new rocks are made. Which part of the rock cycle has happened the most in Australia during recent geological times?

(2 marks)

Answer: Weathering and erosion has occurred the most. The mountains in Australia have been worn down and there are no active volcanoes producing new rocks.

- 19 Place each of these rocks in the appropriate column in the following table. (3 marks)
mudstone, basalt, marble, slate, granite, conglomerate

ROCK TYPES

Igneous	Sedimentary	Metamorphic
Basalt	Sandstone	Marble
Granite	Mudstone	Slate

- 20 Much of the Australian outback has a rusty, red colour. Explain why this is important for Australia's export industries and economy.

(2 marks)

Answer: The rusty red colour is from rusted iron compounds in the rock and sediments. Iron ore (haematite) is a very important export for Australia.

- 21 Complete the table by matching each of these four minerals with the product made from them. **(4 marks)**

bauxite, halite, calcite, haematite

MINERAL	PRODUCT
calcite	cement
halite	table salt
haematite	steel, iron
bauxite	aluminium

22. Ichthyosaurs are large marine reptiles that first appeared in the fossil record in the early Triassic period. They became extinct in the late Cretaceous period. **(2 marks)**

a) How many years did this group live on Earth?

Answer: About 200 million years.

b) What era did the Ichthyosaurs live?

Answer: Mesozoic era.

23. Geologists studied three roadside cuttings to look at the sediments and the fossils they contained to try to work out the stratigraphic order of the sediments from oldest to youngest.

a) Identify two sediments from sediments A – J that contain exactly the same fossils. **(1 mark)**

Answer: Accept either A – I, B – J, C – E, or D – F.

b) In Cutting 1 which is the oldest sediment? **(1 mark)**

Answer: Sediment D

c) Out of all the sediments A – J, which is the oldest sediment? **(2 marks)**

Answer: Sediment G

d) Out of all the sediments A – J, which is the youngest sediment? **(2 marks)**

Answer: Sediment H

24. Use the information provided in the following tables to identify the four minerals and the type of rock. **(5 marks)**

Mineral 1 - Answer: Quartz
Mineral 2 - Answer: Muscovite
Mineral 3 - Answer: Biotite
Mineral 4 - Answer: Haematite
Name of rock - Answer: Slate

Some common rock forming minerals